

LINCOLN 8 ENE, NE, USA

WMO: 744420

Lat: 40.8484N

Lon: 96.5651W

Elev: 362

StdP: 97.05

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.8	-23.3	0.5	-18.6	-20.8	0.6	-16.0	5.6	0.8	4.8	-0.1	0.6	N/A	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.6	23.8	33.0	23.8	31.5	23.3	26.1	31.8	25.1	30.5	24.2	29.6	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.4	30.1	23.5	19.1	28.8	22.5	18.0	27.7	83.2	32.0	78.6	30.6	74.8	29.7	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	3.7	3.1	-24.5	37.2	3.1	1.5	-26.8	38.2	-28.6	39.1	-30.3	39.9	-32.6	41.0		
(5)			-25.2	27.6	2.3	0.8	-26.8	28.2	-28.1	28.6	-29.4	29.0	-31.0	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-4.1	-2.5	5.1	11.0	16.8	22.4	23.1	19.1	11.6	4.4	-2.1	(6)	
(7)		DBStd	11.44	6.91	6.89	6.73	5.74	4.87	3.51	3.44	3.29	4.67	5.48	5.78	6.13	(7)
(8)		HDD10.0	1640	437	353	183	56	6	0	0	0	1	46	183	376	(8)
(9)		HDD18.3	3337	696	584	412	227	88	7	2	3	47	218	419	634	(9)
(10)		CDD10.0	1946	0	2	30	87	217	371	451	407	273	96	14	1	(10)
(11)		CDD18.3	603	0	0	2	8	41	127	194	152	69	9	0	0	(11)
(12)		CDH23.3	6228	0	0	26	121	456	1296	2068	1449	694	113	5	0	(12)
(13)		CDH26.7	2429	0	0	4	34	144	478	912	587	242	28	0	0	(13)
(14)	Wind	WSAvg	1.2	1.3	1.4	1.6	1.6	1.3	1.1	0.9	0.9	1.1	1.2	1.2	(14)	
(15)	Precipitation	PrecAvg	768	18	23	39	71	128	116	97	88	72	54	37	26	(15)
(16)		PrecMax	1101	39	55	93	122	266	224	312	186	154	129	84	110	(16)
(17)		PrecMin	495	2	2	0	16	35	10	23	10	11	2	3	1	(17)
(18)		PrecStd	148	9	14	26	24	65	55	62	47	34	35	23	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	19.9	26.2	30.5	33.3	34.9	37.6	36.3	33.7	30.3	23.6	15.5	(19)
(20)			MCWB	7.4	10.4	15.5	18.2	20.2	24.3	22.3	23.4	22.5	18.1	13.5	8.6	(20)
(21)		2%	DB	11.3	14.1	22.9	26.8	30.1	32.9	35.0	33.9	31.7	26.2	19.6	12.1	(21)
(22)			MCWB	5.7	8.4	13.1	16.2	20.0	23.9	24.5	23.7	22.4	16.7	11.9	7.3	(22)
(23)		5%	DB	8.2	10.6	19.5	23.8	28.1	31.2	33.2	31.8	29.8	23.9	16.6	9.4	(23)
(24)			MCWB	3.8	5.5	11.8	14.4	19.1	23.2	24.9	23.4	21.6	15.5	10.2	5.1	(24)
(25)		10%	DB	5.3	7.2	16.2	21.1	25.8	29.6	31.5	30.0	27.4	21.1	13.9	6.7	(25)
(26)			MCWB	2.3	4.0	10.2	13.2	17.9	22.1	24.4	22.8	20.2	13.9	8.5	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	12.5	17.5	18.8	22.9	26.7	27.7	26.8	24.6	20.5	15.2	11.8	(27)
(28)			MCD B	14.2	17.1	25.0	26.4	29.0	32.4	33.2	32.5	31.1	25.2	19.2	14.1	(28)
(29)		2%	WB	5.8	8.9	15.2	17.0	21.5	25.2	26.4	25.4	23.6	18.0	12.9	7.8	(29)
(30)			MCD B	10.3	15.1	20.0	24.0	27.8	31.0	32.2	30.6	29.8	23.4	18.5	11.3	(30)
(31)		5%	WB	4.1	5.9	12.9	15.6	20.5	24.2	25.5	24.3	22.6	16.4	10.8	5.6	(31)
(32)			MCD B	7.4	10.3	18.9	21.2	26.8	29.9	30.9	29.4	27.8	21.3	15.8	9.5	(32)
(33)		10%	WB	2.4	3.9	10.6	14.0	19.2	23.2	24.5	23.4	21.6	14.8	8.9	3.7	(33)
(34)			MCD B	5.1	7.1	16.9	19.2	24.9	28.7	29.8	28.2	26.1	19.3	13.6	6.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.0	11.5	13.1	13.5	12.5	11.7	11.9	12.1	13.4	13.4	12.9	10.9	(35)
(36)			MCD BR	15.5	16.5	17.8	17.6	15.3	12.7	13.0	13.8	14.2	16.7	16.9	15.1	(36)
(37)			MCWBR	10.5	10.7	10.5	9.2	6.7	6.3	5.8	6.2	6.4	8.8	9.7	10.2	(37)
(38)		5% WB	MCD BR	14.2	16.0	16.0	14.1	12.6	11.7	11.1	11.6	11.7	13.7	15.1	14.2	(38)
(39)			MCWBR	9.9	10.6	9.3	8.1	6.5	6.7	6.1	6.1	6.0	8.2	9.4	10.1	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.282	0.307	0.326	0.383	0.411	0.419	0.426	0.422	0.381	0.333	0.297	0.281	(41)	
(42)		taud	2.468	2.416	2.409	2.268	2.261	2.297	2.277	2.291	2.356	2.479	2.505	2.475	(42)	
(43)		Ebn at Noon	886	909	926	888	868	858	848	841	855	867	860	856	(43)	
		Edh at Noon	76	93	105	130	134	130	131	126	110	87	73	70		
(44)	All-Sky Solar Radiation	RadAvg	2.10	3.00	3.92	4.98	5.86	6.55	6.54	5.70	4.67	3.33	2.35	1.77	(44)	
(45)		RadStd	0.18	0.18	0.34	0.34	0.45	0.46	0.41	0.32	0.31	0.37	0.16	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page